SNES hardware summary

A from-the-ground orientation to the Super Nintendo / Super Famicom for codegen and demo work in this repo. Companion references: the complete register map (generated from `platforms/snes/snes_*.h`) and the 65816 CPU reference. Facts are per the sources in SOURCES.md (nocash *fullsnes*, the SNESdev wiki, anomie's docs, Copetti).

At a glance

Part	What
CPU	Ricoh **5A22** — a WDC **65816** core (16-bit 6502 descendant) + DMA, hardware multiply/divide, joypad auto-read
CPU clock	3.58 MHz fast (FastROM / internal regs), 2.68 MHz slow (ROM default), 2.68/1.79 MHz for RAM/slow regions (NTSC)
Address space	24-bit, 16 MB: 256 banks × 64 KB
WRAM	**128 KB** in banks `$7E` – `$7F`; the low 8 KB mirrored at `$0000` – `$1FFF` in banks `$00` – `$3F` / `$80` – `$BF`
Video	dual **PPU1 + PPU2**; 64 KB VRAM, 512 B CGRAM (256 colours), 544 B OAM (128 sprites)
Audio	**SPC700** CPU + **S-DSP** (8 voices) + 64 KB audio RAM — a separate processor behind 4 mailbox ports
Output	NTSC 256×224 @ ~60 Hz (or 512×448 hi-res/interlace); PAL 256×239 @ 50 Hz

CPU and memory

The 65816 boots in **6502 emulation mode** and is switched to **native mode** once at startup (`crt0` does `CLC; XCE`). In native mode the accumulator and index registers are independently 8- or 16-bit (the M and X status bits); `+mos-a16` runs with a 16-bit accumulator. See the 65816 reference for the programming model and instruction set.

```
bank        contents of each 64 KB bank (offset $0000 → $FFFF)
────────    ──────────────────────────────────────────────────────────────
$00–$3F  →  low-RAM mirror ($0000–$1FFF) · I/O ($2100/$4200/$4300) · LoROM $8000+
$40–$7D  →  cartridge ROM
$7E–$7F  →  128 KB Work RAM (WRAM)        ← high WRAM via far ptr or the $2180 port
$80–$BF  →  mirror of $00–$3F             (FastROM-capable)
$C0–$FF  →  cartridge ROM (HiROM)
```

The 24-bit address space is organised as **256 banks of 64 KB**. The system banks (`$00` – `$3F` and their fast mirror `$80` – `$BF`) share a common low layout:

Range (in a system bank)	Contents
`$0000` – `$1FFF`	Low-RAM — a mirror of the first 8 KB of WRAM (`$7E:0000`)
`$2100` – `$213F`	PPU registers
`$2140` – `$2143`	APU I/O ports
`$2180` – `$2183`	WRAM access port
`$4016` – `$4017`	Serial joypad
`$4200` – `$421F`	CPU I/O (interrupts, mul/div, timers, auto-joypad)
`$4300` – `$437F`	DMA / HDMA channels
`$8000` – `$FFFF`	Cartridge ROM

Banks `$7E` – `$7F` are the full **128 KB of WRAM**. Cartridge ROM is mapped by the board: **LoROM** exposes 32 KB per bank at `$8000` – `$FFFF` ; **HiROM** exposes a full 64 KB per bank. This repo's SNES platform is **LoROM** (see `platforms/snes/link.ld`). High WRAM (`$7E2000` + and bank `$7F`) is reachable from a `$00` -bank program either through the WRAM port (`$2180` – `$2183`) or with a 65816 far pointer (the `+mos-a16` far path).

A **FastROM** board + `MEMSEL` ($420D) bit 0 run banks `$80` + at 3.58 MHz instead of 2.68 MHz.

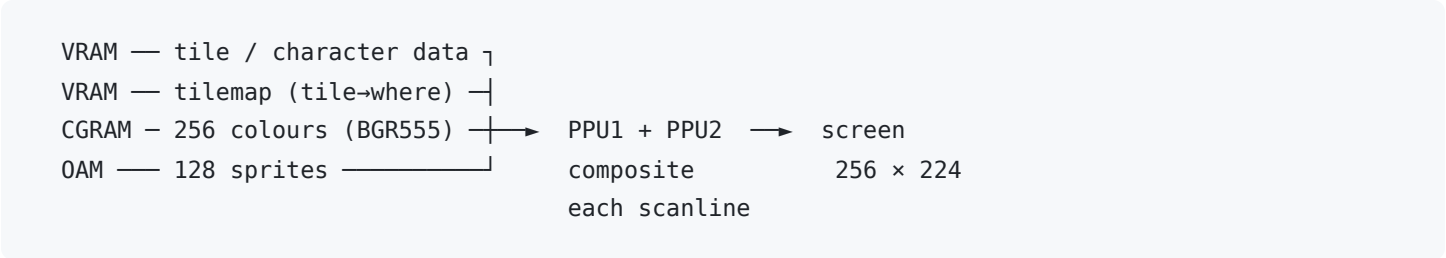
Video (PPU)

The PPU has **no linear framebuffer**. You upload three things into its private memories and it composites them every scanline:

- **VRAM** — 64 KB, **word-addressed** (32 K 16-bit words). Holds tile (character) data and tilemaps. Accessed through `VMADD` / `VMDATA` (`$2116` – `$2119`) with an auto-increment set by `VMAIN` .
- **CGRAM** — 256 palette entries, **BGR555** (`0bbbbbgggggrrrrr` , 15-bit colour). Entry 0 is the backdrop, and pixel value 0 in any tile is transparent.
- **OAM** — 544 B: 128 sprites × 4 B (X, Y, tile, attributes) plus a 32 B high table (X bit 8 + size bit per sprite).

> **The access-window rule (the #1 "nothing shows / garbage shows" bug):** VRAM, CGRAM and OAM are writable **only during force-blank or v-blank**. Writes during active display are dropped. Bring the machine up force-blanked (`INIDISP` bit 7), upload everything, then release the blank last. Also initialise **all** PPU control registers, not just the ones you use — power-on state is indeterminate and some emulators randomise it (`snes_ppu_reset_blank()` in `snes_ppu.h` does this).

How those memories combine into a frame:



Background modes

`BGMODE` ($2105) low three bits select the layer layout / bit depth:

Mode	Layers
0	4 × 2bpp (4-colour) BGs
1	BG1/BG2 4bpp (16-colour) + BG3 2bpp — the common one
2	BG1/BG2 4bpp with per-tile offset
3	BG1 8bpp (256-colour) + BG2 4bpp
4	BG1 8bpp + BG2 2bpp, per-tile offset
5	BG1 4bpp + BG2 2bpp, hi-res 512
6	BG1 4bpp hi-res, per-tile offset

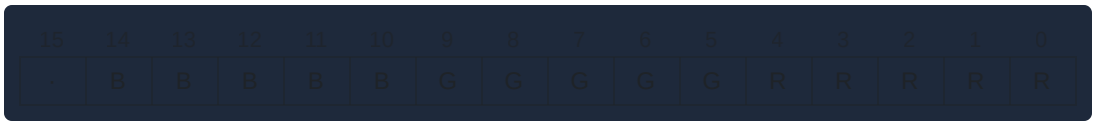
Mode	Layers
7	one 8bpp layer with a full **affine** transform (rotate/scale; pseudo-3D via per-scanline HDMA of the matrix)

bpp = bits/pixel = colours per tile (2bpp = 4, 4bpp = 16, 8bpp = 256). Tiles are 8×8; normal modes store them as **bit-planes interleaved by row pair** (the fiddly part). Mode 7 is the exception — its character data is **linear, 1 byte/pixel** (no bit-planes), at the cost of a 256-tile cap and even/odd VRAM interleaving (tilemap in even bytes, chr in odd). Mode 7 is the natural fit for a per-pixel framebuffer-style image.

Tile (character) data format — 4bpp 8×8 tile = 32 bytes; bit-planes interleaved by row pair, MSB = leftmost pixel:

```
offset  0 : row0 plane0 | row0 plane1   ┐ planes 0+1, rows 0–7  = 16 B
offset  2 : row1 plane0 | row1 plane1   │
     …                                  ┘
offset 16 : row0 plane2 | row0 plane3   ┐ planes 2+3, rows 0–7  = 16 B
     …                                  ┘
a pixel's 4-bit index = (p3 p2 p1 p0) read down the planes at that column
2bpp = 16 B (planes 0+1) · 8bpp = 64 B (planes 0–7) · Mode 7 chr = linear 1 byte/pixel
```

Colour format — each CGRAM entry is 16-bit **BGR555** (bit 15 ignored); `SNES_RGB(r,g,b)` packs it:



`·` =unused `B` =blue[14-10] `G` =green[9-5] `R` =red[4-0]

Sprites, windows, colour math

Up to **128 sprites**, ~32 per scanline, sizes 8×8 to 64×64 (a pair selectable via `OBSEL`). Two **window** regions can mask any layer (`W12SEL` / `W34SEL` / `WOBJSEL` , `WH0` – `WH3`), and a **colour-math** unit adds/subtracts a fixed colour or the sub-screen (`CGWSEL` / `CGADSUB` / `COLDATA`) for transparency and lighting effects.

DMA / HDMA

Eight channels (`$4300` – `$437F`). **General-purpose DMA** (triggered by `MDMAEN` $420B) blasts up to 64 KB CPU→PPU (or back) while the CPU stalls — the only practical way to fill VRAM in one v-blank. **HDMA** (`HDMAEN` $420C) streams a few bytes to registers each scanline, for gradients, window animation, and Mode-7 matrix updates. Same access-window rule applies: do GP-DMA into VRAM/CGRAM/OAM in force-blank or v-blank. See the DMA / HDMA section of the register map.

```
A-bus source              DMA channel x           B-bus → PPU port
A1Tx:A1Bx (ROM/RAM) ──►  DMAPx · BBADx  ──────►  VMDATA / CGDATA / OAMDATA
      └ DASx bytes ┘            ▲
                                └─ write MDMAEN bit x   (starts it; CPU stalls)
```

Audio

The audio subsystem is a **physically separate computer**: an **SPC700** CPU with its own 64 KB RAM and an **S-DSP** (8 stereo voices, ADSR, echo). The 65816 cannot address any of it directly — all communication is a handshake through the four **APU I/O ports** (`$2140` – `$2143`) with the SPC700 boot ROM. Its internal registers are therefore out of scope for the CPU-visible register map.

Timing and interrupts

One NTSC frame is ~224 visible scanlines + ~38 of v-blank, ~1/60 s. The **v-blank** window is the per-frame budget for VRAM/CGRAM/OAM writes (a few KB by hand, far more via DMA). Enable the **v-blank NMI** with `NMITIMEN` ($4200) bit 7; an optional **H/V IRQ** fires at a programmed `HTIME` / `VTIME` position. The usual loop: compute into a RAM shadow during active display, then in the NMI handler DMA the changed bytes to the PPU during v-blank. A purely static image can instead just force-blank, build everything once, and release the blank — the simplest thing that displays.

```
one NTSC frame ≈ 262 scanlines ≈ 1/60 s
line   0 ┃ ▓▓▓ visible display (~224 lines) ▓▓▓   VRAM/CGRAM/OAM writes: DROPPED
line 224 ┃ ── v-blank begins ──────────────────   NMI fires (NMITIMEN bit 7)
         ┃ ███ v-blank (~38 lines) ███           VRAM/CGRAM/OAM + DMA: SAFE
line 262 ┃ (next frame)
loop: compute into a RAM shadow during display → DMA it to the PPU during v-blank
```